

CURRICULUM VITAE

Abhirup Datta

PROFESSIONAL DATA

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Twitter: @dattascience

EDUCATION AND TRAINING

Degree	Year	Institution
PhD, Biostatistics	2016	University of Minnesota, Twin Cities, MN
MStat (Specialization in Math-Stat-Probability)	2010	Indian Statistical Institute, Kolkata, India
BStat (with Honors)	2008	Indian Statistical Institute, Kolkata, India,

PROFESSIONAL EXPERIENCE

Johns Hopkins University

- 2022 – present: Associate Professor, Department of Biostatistics, Johns Hopkins University.
- 2016 – 2021: Assistant Professor, Department of Biostatistics, Johns Hopkins University.
- 2017 – present: Affiliate Faculty, The Spatial Science for Public Health Center, Johns Hopkins University.

Other Non-JHU Professional Experience

- 2010 – 2012: Quantitative analyst, Morgan Stanley, New York.

HONORS AND AWARDS

- (2024) Committee of Presidents of Statistical Societies (COPSS) Emerging Leader Award. This award recognizes statistical scientists within 10 years of terminal-degree who show evidence of and potential for leadership and who will help shape and strengthen the field
- (2023) Abdel El-Shaarawi Early Investigator's Award, The International Environmetrics Society (TIES). This award is given annually to an researcher within 12 years of completion of their PhD for their statistical contributions to Environmental Sciences.
- (2021) Early Career Award in Statistics and Data Sciences (ECADS), International Indian Statistical Association (IISA). This annual award recognizes most outstanding contributions in Applications and Statistical Practice by a person under 41 years of age.
- (2021) Early Investigator Award (EIA), American Statistical Association Section on Statistics and the Environment (ENVR). This annual award recognizes outstanding contributions to environmental statistics in the first 12 years after obtaining terminal degree.
- (2018) Honorable mention (Finalist): Savage Award (Applied Methodology), International Society for Bayesian Analysis (ISBA).
- (2017) ASA Outstanding Statistical Application Award, American Statistical Association.
- (2016) ENAR Distinguished Student Paper Award, International Biometric Society, Austin, TX.
- (2016) Delta Omega Honorary Society Student Inductee (Pi Chapter), Minneapolis, MN.
- (2014) JSM Student Paper Award, American Statistical Association, Section on Bayesian Statistical Science, Boston, MA. Also selected for best paper award in Statistics and the Environment Section
- (2014) Best Paper Award, Division of Biostatistics, University of Minnesota, Minneapolis, MN.
- (2016) Best Student Seminar Presentation Award, Division of Biostatistics, University of Minnesota, Minneapolis, MN.
- (2015) Inter-disciplinary Doctoral Fellowship 2015-16, Division of Biostatistics, University of Minnesota Graduate School, Minneapolis, MN.

Teaching, Advising, and Mentoring awards:

- (2020) JHU AMTRA Award (Advising, Mentoring, & Teaching Recognition) by JHSPH Student Assembly for 2019-2020.
- Excellence in Teaching awards, Johns Hopkins Bloomberg School of Public Health,
 - (1) (2023, Term 4) Probability Theory IV
 - (2) (2023, Term 3) Probability Theory III
 - (3) (2022) Probability Theory IV
 - (4) (2021) Probability Theory IV
 - (5) (2018) Probability Theory IV
- (2014) Outstanding Teaching Assistant Award, Division of Biostatistics, University of Minnesota, Minneapolis, MN.

Peer review awards:

- (2018) Top 1% of reviewers in Mathematics, Publons.
- (2017) Top 1% of reviewers in Mathematics, Publons.

RESEARCH GRANT AWARDS AS PRINCIPAL INVESTIGATOR

- **NIEHS R01: Statistical methods for air-pollution studies using low-cost monitors**
The National Institute of Environmental Health Sciences (NIEHS) R01ES033739
Dates: February 2022 to November 2026.
Principal Investigators: Abhirup Datta.
Award amount: \$1,326,376.00
Responsibility: Principal Investigator.
- **NIBIB R01: Individualized spatial topology in functional neuroimaging**
National Institute Of Biomedical Imaging And Bioengineering (NIBIB) R01EB026549
Dates: April 2024 to March 2028.
Multiple Principal Investigators (M-PI): Abhirup Datta, Martin Lindquist (PI), Tor Wager
Award amount: \$ 2,196,809.00
Responsibility: Multiple Principal Investigator (M-PI) .
- **Broadening the applicability of minimally-invasive-tissue-sampling (MITS)-based verbal autopsy (VA) calibration to improve global mortality estimates**
Bill & Melinda Gates Foundation
Dates: December 2021 to April 2024.
Principal Investigators: Abhirup Datta.
Award amount: \$1,099,940.00
Responsibility: Principal Investigator.
- **NSF-DMS award: Highly multivariate geo-statistics using graphical models**
National Science Foundation Division of Mathematical Sciences (NSF-DMS) award DMS-1915803
Dates: July 2019 to June 2023.
Principal Investigators: Abhirup Datta.
Award amount: \$180,000
Responsibility: Principal Investigator.
- **Air Pollution and COPD Hospitalizations in Baltimore in the Context of COVID-19**
Alliance for a Healthier World COVID-19 Launchpad Grant
Dates: June 2020 to May 2021.
Principal Investigators: Kirsten Koehler and Abhirup Datta.
Responsibility: Co-Principal Investigator.
- **NIDA R21: Improved Heritability Estimation by Spatial Mapping of Genetic Relationships**
National Institute on Drug Abuse (NIDA) R21DA046188
Dates: July 2018 to Jun 2020.
Principal Investigator: Saonli Basu.
Responsibility: Principal investigator on sub-contract.
- **Statistical Maps of Air Quality in Baltimore City Using Low-Cost Monitoring Data**
(Bloomberg American Health Initiative Spark Award)
Dates: July 2018 to June 2019.
Principal Investigators: Abhirup Datta and Kirsten Koehler.
Award amount: \$71,000
Responsibility: Principal Investigator.

PUBLICATIONS

The white numbers in black boxes indicates manuscripts where Dr. Datta is the first author or senior/corresponding author;

* indicates a mentored student or post-doctoral fellow of Dr. Datta;

† indicates equal contributions.

Highlights

- 1 *Zhan W, **Datta A** (2024) Neural networks for geospatial data *Journal of the American Statistical Association Theory and Methods (In Press)* [DOI](#)
- 2 * Heffernan C, Koehler K, Levy-Zamora M, Buehler C, Gentner D, Peng RD, **Datta, A** (2024) A machine learning based interrupted time series framework for studying causal changes in air pollution due to policy interventions: A case study in COVID-19 lockdowns *American Journal of Epidemiology (In Press)*
 - Student paper award for C. Heffernan from the American Statistical Association Section on Statistics and the Environment at the Joint Statistical Meetings (2023)
- 3 *Heffernan C, Peng RD, Gentner D, Koehler K, **Datta, A** (2023) A dynamic spatial filtering approach to mitigate underestimation bias in field calibrated low-cost sensor air-pollution data *Annals of Applied Statistics* vol. 17, no. 4, pp. 3056–3087, 2023.
 - Student paper award for C. Heffernan at the EnviBayes 2023 conference of the International Society for Bayesian Analysis (ISBA).
- 4 *Saha A, Basu S, **Datta A** (2023) Random forests for spatially dependent data *Journal of the American Statistical Association Theory and Methods* 118.541 665–683.
- 5 Wang G, **Datta A**, Lindquist M (2023) Improved fMRI-based Pain Prediction using Bayesian Group-wise Functional Registration *Biostatistics (Accepted)* [ArXiv](#)
- 6 *Dey D, **Datta A**, Banerjee S (2022) Graphical Gaussian Processes for highly multivariate spatial data *Biometrika*, 109(4), 993–1014.
 - Student paper award for D. Dey from the American Statistical Association Section on Bayesian Statistics (SBSS) at the Joint Statistical Meetings (2021).
- 7 Levy-Zamora M, Buehler C, **Datta, A**, Gentner D, Koehler K (2023) Identifying optimal co-location calibration periods for low-cost sensors *Atmospheric Measurement Techniques* 16.1: 169–179.
- 8 Weber L, *Saha A, **Datta A**, Hansen K, Hicks S (2023) nnSVG: scalable identification of spatially variable genes using nearest-neighbor Gaussian processes *Nature Communications* 14.1: 4059.
- 9 *Fiksel J, **Datta A**, Amouzou A, Zeger S. (2022) Generalized Bayes Quantification Learning under Dataset Shift *Journal of the American Statistical Association Theory and Methods*, 117(540), 2163–2181.
- 10 Finley AO, **Datta A**, Banerjee S. (2022) spNNGP R package for Nearest Neighbor Gaussian Process models *Journal of Statistical Software*, 103(1), 1–40.
- 11 *Fiksel J, Zeger S, **Datta A** (2021) A Transformation-free Linear Regression for Compositional Outcomes and Predictors *Biometrics*, 78, 974– 987.

- 12 Wang G, **Datta A**, Lindquist M (2021) Bayesian Functional Registration of fMRI Data *Annals of Applied Statistics* , 16(3), 1676-1699.
- 13 **Datta A**, *Fiksel J, Amouzou A, Zeger S. (2020) Regularized Bayesian transfer learning for population level etiologic distributions *Biostatistics*, ISSN 1465-4644
- 14 **Datta A**, *Pita, A, Rao, A, Sithole, B, Mnisi, Z, and Baral, S. (2020) Size Estimation of Key Populations in the HIV Epidemic in eSwatini using incomplete and misaligned capture-recapture data *Annals of Applied Statistics*, 14(3), 1207-1241
- 15 [†] Butler EE, [†] **Datta A**. / ... 48 authors ... / Reich, PB. (2017) Mapping local and global variability in plant trait distributions *Proceedings of the National Academy of Sciences* 114(51): E10937-E10946
- 16 **Datta A**, Zou H. (2017) CoCoLasso for High-dimensional Error-in-variables Regression *Annals of Statistics* 45(6): 2400-2426
- 17 **Datta A**, Banerjee S, Finley AO, Hamm NAS, Schaap M. (2016) Non-separable Dynamic Nearest Neighbor Gaussian Process Models for Large Spatio-temporal Data with Application to Particulate Matter Analysis *Annals of Applied Statistics* 10(3): 1286-1316
 - American Statistical Association Outstanding Statistical Application award (2017).
 - Eastern North American Region (ENAR) distinguished student paper award for A. Datta (2016).
- 18 **Datta A**, Banerjee S, Finley AO, Gelfand AE. (2016) Hierarchical Nearest Neighbor Gaussian Process models for Large Geostatistical Datasets *Journal of the American Statistical Association Theory and Methods* 111(514) 800-812
 - One of top 5 most cited papers in the Journal of the American Statistical Association between 2016-2020.
 - American Statistical Association Section on Bayesian Statistics (SBSS) student paper award for A. Datta at the Joint Statistical Meetings (2014).

Statistical Methods

- 19 Bonas M, **Datta A**, Wikle CK, Boone EL, Alamri FS, Hari BV, Kavila I, Simmons SJ, Jarvis SM, Burr WS, Pagendam D, Chang W, Castruccio S (2024) Assessing Predictability of Environmental Processes with Statistical and Machine Learning Models *Environmetrics*
 - Selected as a discussion paper for Environmetrics.
- 20 *Dey D, **Datta A**, Banerjee S (2023) Modeling Multivariate Spatial Dependencies Using Graphical Models *New England Journal of Statistics and Data Science (Accepted)*
- 21 **Datta A**. (2023) Invited Discussion of "Saving Storage in Climate Ensembles: A Model-Based Stochastic Approach" *Journal of Agricultural Biological and Environmental Statistics* 28.2 (2023): 352-357.
- 22 Wikle CK, **Datta A**, Hari BV, Boone EL, Sahoo I, Kavila I, Castruccio S, Simmons SJ, Burr WS, Chang W (2022) An Illustration of Model Agnostic Explainability Methods Applied to Environmental Data *Environmetrics*, 34(1), e2772.

- 23 *Saha A, **Datta A**, Banerjee S Scalable Predictions for Spatial Probit Linear Mixed Models Using Nearest Neighbor Gaussian Processes *Journal of Data Science*, 20(4), 533-544,
- 24 *Saha A, Basu S, and **Datta A.**, (2022). RandomForestsGLS: An R package for Random Forests for dependent data *Journal of Open Source Software*, 7(70), 3780
- 25 Gao L., **Datta A**, Banerjee S (2022) Hierarchical Multivariate Directed Acyclic Graph Auto-Regressive (MDAGAR) Models for Spatial Diseases Mapping *Statistics in Medicine*, 41(16): 3057–3075
- 26 **Datta A** (2021) Sparse nearest-neighbor Cholesky matrices in spatial statistics *Wiley Interdisciplinary Reviews: Computational Statistics*, e1574.
- 27 **Datta A**, Zou H. (2019) A note on cross-validation for Lasso under measurement errors *Technometrics*, 62(4), 549–556
- 28 **Datta A**, Banerjee S, Hodges JS., Gao, L. (2019) Spatial disease mapping using Directed Acyclic Graph Auto-Regressive (DAGAR) models *Bayesian Analysis* 14(4), 1221–1244
- 29 Gao, L., **Datta A**, Banerjee S, (2020) Spatial Modeling for Correlated Cancers Using Bivariate Directed Graphs *Annals of Cancer Epidemiology* 4, ISSN 2616-4213
- 30 Taylor-Rodriguez D, Finley AO, **Datta A**, Babcock C, Andersen H, Cook BD, Morton DC, Banerjee S. (2019) Spatial Factor Models for High-Dimensional, Large Spatial Data: An Application in Forest Variable Mapping *Statistica Sinica* 26(29) 1155–1180
- 31 Zhang L, **Datta A**, Banerjee S. (2019) Practical Bayesian Inference for Massive Spatial Data on Modest Computing Environments *Statistical Analysis and Data Mining: The ASA Data Science Journal* 12.3:197-209.
- 32 Finley AO, **Datta A**, Cook BC, Morton DC, Andersen HE, Banerjee S. (2019) Efficient algorithms for Bayesian Nearest Neighbor Gaussian Processes *Journal of Computational and Graphical Statistics* 28.2 (2019): 401-414.
- 33 Heaton MJ, **Datta A**, Finley AO, Furrer R, Guhaniyogi R, Gerber F, Gramacy RB, Hammerling D, Katzfuss M, Lindgren F, Nychka DW, Sun F, Zammit-Mangion A. (2019) A Case Study Competition Among Methods for Analyzing Large Spatial Data *Journal of Agricultural, Biological and Environmental Statistics* 24(3) 398–425.
 - Best Paper award for 2018-2019 in the Journal of Agricultural, Biological and Environmental Statistics by the International Biometric Society
- 34 **Datta A**, Lin W, Rao A, Diouf D, Kouame A, Edwards JK, Bao L, Louis TA, Baral SB (2019) Bayesian estimation of MSM population in Côte d’Ivoire *Statistics and Public Policy* 6(1), 1-13.
- 35 **Datta A**, Zou H, Banerjee S. (2019) Bayesian high-dimensional regression for change point analysis *Statistics and Its Interface* 12(2), 253-264.
- 36 *Saha A, **Datta A.** (2018) BRISC: Bootstrap for rapid inference on spatial covariances *Stat e184*
 - American Statistical Association Section on Statistical Computing Student paper award for A. Saha at Joint Statistical Meetings, 2018.

- One of two papers selected for ‘Highlights of the Stat journal’ session at International Statistical Institute World Congress, 2019.

37 **Datta A**, Banerjee S, Finley AO, Gelfand AE. (2016) On nearest-neighbor Gaussian process models for massive spatial data *Wiley Interdisciplinary Reviews: Computational Statistics* 8(5) 162-171

Scientific Applications

- 38** *Fiksel J, *Gilbert B, Wilson E, Kalter H, Kante A, Akum A, Blau D, Bassat Q, Macicame I, Gudo E, Black R, Zeger S, Amouzou A, **Datta A** (2023) Correcting for verbal autopsy misclassification bias in cause-specific mortality estimates *American Journal of Tropical Medicine and Hygiene* 108.5 Suppl: 66.
- 39** Wilks M, Green T, Rule AM, Zamora ML, Buehler C, **Datta A**, Gentner DR, Putcha N, Hansel NN, Kirk GD, Raju S, McCormack M, Koehler K (2023) Evaluation of Calibration Approaches for Indoor Deployments of PurpleAir Monitors *Atmospheric Environment* 119944.
- 40** Rosenblum, M, Chin, ET, Ogburn, EL, Nishimura, A, Westreich, D, **Datta, A**, Vanderplas, S, Cuellar, M and Thompson, WC (2023) Misuse of Statistical Method Results in Highly Biased Interpretation of Forensic Evidence in Guyll et al. *Law., Probability, and Risk* (Accepted)
- 41** *Gilbert B, *Fiksel J, Wilson E, Kalter H, Kante A, Akum A, Blau D, Bassat Q, Macicame I, Gudo E, Black R, Zeger S, Amouzou A, **Datta A** (2023) Multi-cause calibration of verbal autopsy-based cause-specific mortality estimates of children and neonates in Mozambique *American Journal of Tropical Medicine and Hygiene* 108.5 Suppl: 78.
- 42** Lin J, Buehler C, **Datta A**, Gentner D, Koehler K, Levy-Zamora M (2023) Laboratory and Field Evaluation of a Low-cost Methane Sensor and Key Environmental Factors for Sensor Calibration *Environmental Science: Atmospheres* 3.4: 683-694.
- 43** Ivalda M, Almamy K, Wilson E, *Gilbert B, Nhachungue S, Monjane C, Adriano A, Chicumbe S, Jani, I, Kalter H, **Datta A**, Zeger S, Black R, Samo G, Amouzou A (2023) Countrywide Mortality Surveillance for Action – COMSA - in Mozambique: Results from a national sample vital registration system for mortality and cause of death *American Journal of Tropical Medicine and Hygiene* 108.5 Suppl: 5.
- 44** Seal S, **Datta A**, Basu S (2022) Efficient Estimation of SNP Heritability using Gaussian Predictive Process in Large scale Cohort Studies *PLOS Genetics*, 18(4): e1010151.
- 45** Patton AN, **Datta A**, Levy-Zamora M, Buehler C, Xiong F, Gentner D, Koehler K (2022) Non-linear probabilistic calibration of low-cost environmental air pollution sensor networks for neighborhood level spatiotemporal exposure assessment *Journal of Exposure Science and Environmental Epidemiology* 32.6: 908-916.
- 46** Zamora ML, Buehler C, Lei H, **Datta A**, Xiong F, Gentner D, Koehler, K (2022) Evaluating the performance of using low-cost sensors to calibrate for cross-sensitivities in a multipollutant network *Environmental Science and Technology Engineering* 2.5: 780-793.

- 47 Butler EE, Wythers KR, Flores-Moreno, H, Ricciuto DM, **Datta A**, Banerjee A, Atkin OK, Kattge J, Thorton PE, Madhur A, Burrascano S, Byun C, Cornelissen JHC, Forey E, Jansen S, Kramer K, Minden V, and Reich PB (2022) Increasing functional diversity in a global land surface model illustrates uncertainties related to parameter simplification *Journal of Geophysical Research - Biogeosciences* 127.3 (2022): e2021JG006606.
- 48 Butler EE, Wythers KR, Flores-Moreno, H, Chen M, **Datta A**, Ricciuto DE, Atkin OK, Kattge J, Thorton PM, Banerjee A, Reich PB (2021) Updated respiration routines alter spatio-temporal patterns of carbon cycling in a global land surface model *Environmental Research Letters* 16(10) p. 104015.
- 49 **Datta, A**, *Saha, A, Levy-Zamora, M, Buehler, Colby, Hao, L, Xiong, F, Gentner DR, Koehler K (2020) Statistical field calibration of a low-cost PM2.5 monitoring network in Baltimore *Atmospheric Environment* 242, 117761, ISSN 1352-2310
- 50 Flores-Moreno H, Fazayeli F, Banerjee A, **Datta A**, Kattge J, Butler EE, Atkin O, Wythers K, Chen M, Anand M, Bahn M, Burrascano S, Byun C, Cornelissen J, Craine J, Gonzalez-Melo A, Hatttingh W, Jansen S, Kraft N, Kramer K, Laughlin D, Minden V, Niinemets U, Onipchenko V, Penuelas J, Soudzilovskaia N, Reich PB. (2019) Robustness of trait connections between multiple plant organs across environmental gradients, growth forms *Global Ecology and Biogeography* 28(12), 1806–1826
- 51 Edwards JK, Hileman S, Donastorg Y, Sanchez R, Zadrozny S, Baral SB, Hargreaves J, Fearon E, Zhao J, **Datta A**, Weir SS. (2018) Estimating sizes of key populations at the national level: considerations for study design, analysis *Epidemiology* 29(6): 795–803

Manuscripts invited for Revision/Resubmission

- 52 *Gilbert B, Ogburn EL, **Datta A** Consistency of common spatial estimators under spatial confounding (Invited for resubmission, Biometrika) [ArXiV](#)
- 53 *Gilbert B, **Datta A** Visibility graph-based covariance functions for scalable spatial analysis in non-convex domains (Major Revision, Biometrics) [ArXiV](#)
- 54 Pramanik S, Zeger S, Blau D, **Datta A** Modeling Structure and Country-specific Heterogeneity in Misclassification Matrices of Verbal Autopsy-based Cause of Death Classifiers (Major Revision, Annals of Applied Statistics) [ArXiV](#)
- 55 Tang B, Zhao Y, Caffo B, **Datta A** Direct Bayesian Regression for Distribution-valued Covariates (Major Revision, Electronic Journal of Statistics) [ArXiV](#)
- 56 *Dey D, **Datta A**, Banerjee S Graph-constrained Analysis for Multivariate Functional Data using Graphical Gaussian Processes (Major revision, Journal of multivariate analysis) [ArXiV](#)
- 57 *Gilbert B, **Datta A**, Casey JA, Ogburn EL A causal inference framework for spatial confounding (Revision, Statistical Science) [ArXiV](#)
- 58 *Saha A, **Datta A** Random forests for binary geospatial data (Invited for resubmission, Biometrika) [ArXiV](#)

Manuscripts Submitted

59 Zhao Y, **Datta A**, Tang B, Zipunnikov V, Caffo B Density-on-Density Regression [ArXiv](#)

Published Open-access Software

1 **GeospaNN (2024)**

GeospaNN is a Python package for using neural networks for geospatial data, available at <https://pypi.org/project/geospaNN/>. GeospaNN implements the NNGLS method of Zhan and Datta (2024, JASA). NNGLS is a graph neural network (GNN) for fast estimation and spatial prediction (kriging) from large geospatial datasets, by combining multi-layer perceptrons, Gaussian processes, and the generalized least squares (GLS) loss.

2 **BRISC (2018) (36,000+ CRAN downloads as of Jan, 2024)**

BRISC is an R-package on CRAN for rapid estimation, prediction and inference for large spatial data in a frequentist setup. BRISC estimation and prediction relies on nearest neighbor approximations of the spatial Gaussian Process likelihood, and uses a scalable parametric bootstrap to provide inference for all spatial parameters. To our knowledge, currently BRISC is the only R-package that provides confidence intervals in a frequentist setup for all parameters including the spatial variance and range of Gaussian Process. Inference from BRISC is highly competitive with those obtained on Bayesian approaches relying on MCMC, while being manifold times faster.

3 **spNNGP (2017) (27,000+ CRAN downloads as of Jan 2024)**

spNNGP is an R package on CRAN for fully Bayesian analysis of massive spatial data. Spatial analysis of point process data is usually computationally expensive requiring memory and computations that are quadratic and cubic in the number of locations where data is observed. spNNGP implements a class of scalable Nearest Neighbor Gaussian Process models that uses memory and computations that are linear in the size of the data. spNNGP enables fast fully Bayesian inference of all parameters and proper uncertainty quantified predictions at new locations. An MCMC-free hybrid Bayesian conjugate NNGP is also included which is super fast even for spatial datasets with millions of locations. The new version of spNNGP also has the option to run Bayesian spatial GLM for binary spatial data using Nearest Neighbor Gaussian Processes.

4 **codalm (2020) (21000+ CRAN downloads as of Jan 2024)**

codalm is an R-package for linear modeling of compositional data (coda). It implements a simple transformation-free regression of a compositional outcome on a compositional prediction using an M-estimation method. Estimates of the regression-coefficient matrix, bootstrap-based confidence intervals are provided. A permutation based test of linear association is also offered.

5 **RandomForestsGLS (2021) (18000+ CRAN downloads as of Jan 2024)**

RandomForestsGLS is an R-package for fitting non-linear regression models on dependent data (spatial and temporal) with Generalised Least Square (GLS) based Random Forests (RF-GLS) detailed in Saha, Basu and Datta (2020). For spatial data, 'RandomForestsGLS' combines the strengths of Random Forest and Gaussian Process to estimate and predict non-linear functions using nearest neighbor Gaussian Process. For time-series data, 'RandomForestsGLS' uses the AR (auto-regressive) process covariance structure with Random Forests for estimation.

6 **calibratedVA (2018) (Github download stats not available)**

calibratedVA is an R-package on Github for local calibration of national and sub-national cause specific mortality (CSMF) estimates produced by algorithms based on verbal autopsy data. These computer coded verbal autopsy (CCVA) algorithms usually rely on non-local gold standard training data and can be inaccurate in a local context. calibratedVA uses the output of the CCVA algorithm and limited amount of local gold standard data to update the CSMF estimates using a fast Bayesian hierarchical model. calibratedVA also has an ensemble calibration option where

outputs from multiple CCVA algorithms are used to produce an unified calibrated CSMF estimate. the package can also be used in other general contexts to calibrate any discrete classifier (or a set of classifiers) based on limited local labeled data.

MENTORING

PhD Advisees

Current:

- 1 Zhang, Wentao, Doctor of Philosophy, Biostatistics (2020 – present, co-advised with Hongkai Ji)
- 2 Song, Jiafang, Doctor of Philosophy, Biostatistics (2021 – present)
- 3 Anik Burman, Doctor of Philosophy, Biostatistics (2022 – present)

Graduated:

- 4 Heffernan, Claire, Doctor of Philosophy, Biostatistics (2019 – 2024)
 - Next position Merck.
- 5 Gilbert, Brian, Doctor of Philosophy, Biostatistics (2019 – 2023, co-advised with Betsy Ogburn).
 - Currently Postdoctoral fellow at the New York University (NYU)
- 6 Dey, Debangana, Doctor of Philosophy, Biostatistics (2017 – 2022, co-advised with Vadim Zipunnikov)
 - Currently Postdoctoral fellow at the National Institute of Mental Health (NIMH)
- 7 Saha, Arkajyoti, Doctor of Philosophy, Biostatistics (2016 – 2021, co-advised with Nilanjan Chatterjee).
 - Incoming **Tenure-track Assistant Professor** for 2024 at Department of Statistics, University of California Irvine
 - Currently Postdoctoral fellow at the Department of Statistics at University of Washington
- 8 Fiksel, Jacob, Doctor of Philosophy, Biostatistics (2015 – 2020).
 - Currently at Vertex Pharmaceuticals, Boston, MA

ScM Advisees

- 9 Lin, Yi-Ting, Master of Science, Biostatistics (2022 – 2023)
 - Currently PhD student at Department of Biostatistics, University of Michigan
- 10 Xiang, Chen, Master of Science, Biostatistics (2020 – 2021)
 - Currently PhD student at Department of Biostatistics, University of California Los Angeles
- 11 Pita, Andrew, Master of Science, Biostatistics (2017 – 2019)

Post-doctoral fellows

- 12 Dr. Sandipan Pramanik (2022 -)
- 13 Bora Jin (August, 2023 -)

JHU Diversity Summer Internship Program (DSIP) Mentorship

- 14 Hall, Byron, Intern (Summer 2023)
- 15 Thomas, Bella, Intern (Summer 2023)
- 16 Griffin, Karen, Intern (Summer 2022)

Other mentored students

- 17 Jungin Choi (PhD student, 2023 - , Biostatistics, co-advisor: Martin Lindquist)
- 18 Dr. Bohao Tang (PhD, 2023, Biostatistics, advisor: Brian Caffo)
- 19 Dr. Guoqing Wang (PhD, 2022, Biostatistics, advisor: Martin Lindquist)
- 20 Wenyi Lin (ScM, 2017, Biostatistics, advisor: Scott Zeger)

TEACHING

Classroom Instruction - Principal Instructor (JHSPH)

- 2018-2023, 140.724 Probability Theory IV (4th term, 6 years).
- 2023, 140.724 Probability Theory III (3rd term).
- 2021, 140.850 Advanced Spatial Statistics (3rd term).
- 2019, Biostatistics PhD seminar (3rd term).
- 2018, 140.850 Advanced spatial statistics (3rd term).
- 2017, 140.850 Scalable methods for large spatial data (4th term).

Short Courses

- Invited short course on Statistical and machine learning for spatial data, International Biometric Society Meeting (2024, scheduled).
- Full day invited short course on Bayesian models for high dimensional spatial data, Joint Statistical Meetings (2017).

ACADEMIC SERVICE

Scientific Advisory Committees

2023 - Pathology-informed Reference Death Archive for the World Health Organization

Program Leadership

2023 - Program chair-elect 2024 for the American Statistical Association Section on the Statistics and the Environment

Editorial Boards

2025 - Journal of the American Statistical Association
2023 - Journal of the Royal Statistical Society Series B
2023 - Journal of the Royal Statistical Society Series A
2022 - Biometrics
2022 - Sankhya (Series B)
2020 - Journal of Computation and Graphical Statistics

Peer Review Activities

1. Advances in Statistical Climatology Meteorology and Oceanography
2. American Journal of Epidemiology (AJE)
3. Artificial Intelligence and Statistics Conference 2021 (AISTATS 2021)
4. Annals of Applied Statistics
5. Annals of Statistics
6. Bayesian Analysis
7. Biometrics
8. Biometrika
9. Biostatistics
10. Brazilian Journal of Probability and Statistics
11. Canadian Journal of Statistics
12. Computational Statistics and Data Analysis (CSDA)
13. Electronic Journal of Statistics
14. Environmental Health Policy (EHP)

15. Environmental Science and Technology (ES&T)
16. Environmetrics
17. Harvard Data Science Review
18. IEEE Transactions on Pattern Analysis and Machine Intelligence
19. Journal of Agricultural Biological and Environmental Statistics (JABES)
20. Journal of the American Statistical Association Applications and Case Studies (JASA-ACS)
21. Journal of the American Statistical Association Theory and Methods (JASA-TM)
22. Journal of Computation and Graphical Statistics (JCGS)
23. Journal of Multivariate Analysis (JMVA)
24. Journal of the Royal Statistical Society Series B (JRSSB)
25. Journal of the Royal Statistical Society Series C (JRSSC)
26. New England Journal of Statistics and Data Science (NEJSDS)
27. Sankhya A
28. Scientific Reports
29. Spatial Statistics
30. Statistica Sinica
31. Statistics and Computing
32. Statistics in Medicine

Johns Hopkins Bloomberg School of Public Health

- Elected member of the Faculty Senate (2021 – 2023)

Department of Biostatistics, Johns Hopkins University

- Honors and Awards committee (2021 – present)
- Co-leader of the Bayesian Learning and Spatio-temporal (BLAST) modeling working group, (2020 – present)
- Graduate students admissions committee (2019 - present)
- Curriculum committee, Biostatistics Retreat (2018)
- Co-leader of the Spatial Statistics and Small area estimation (SAESS) working group, (2016 – 2018)
- Faculty Recruitment Committee (2017)
- Organizer, Biostatistics departmental seminars (2017)

External Letters of support for Promotion or Tenure

George Mason University

Grant Review Panels

- (2023) Faculty Innovation Fund Panel, Johns Hopkins Bloomberg School of Public Health
- (2020) National Science Foundation (NSF) Division of Mathematical Sciences (DMS)
- (2023) National Science Foundation (NSF) Methodology, Measurement, and Statistics (MMS) Program

Program Development

- 2023 Eastern North American Region (ENAR) conference Program Committee
- Ad-hoc committee of the International Society for Bayesian Analysis (ISBA) for early career awards (2021)
- Scientific Program Committee for the International Indian Statistical Association conference (IISA) (2021).

Professional Memberships

- American Statistical Association
- International Biometric Society (Eastern North American Region (ENAR))
- International Indian Statistical Association (IISA)
- The International Environmetric Society (TIES) of the International Statistical Institute (ISI)
- International Society for Bayesian Analysis (ISBA)

Scientific Communication

PLENARY AND KEYNOTE LECTURES

- 1 July 2024 (upcoming) Keynote Lecture, 10th International Workshop on Spatio-Temporal Modeling (METMA XI), Lancaster, UK
- 2 July 2023 Plenary Lecture, The International Environmetrics Society Meeting, Peterborough, Canada

INVITED PANELS

- 3 July, 2024 (upcoming) – Panelist at Grants360 Discussion of funding from the National Science Foundation (NSF), Johns Hopkins Bloomberg School of Public Health
- 4 May, 2024 – Panelist at Conference on Evaluating the Science of Geospatial AI, Harvard University
- 5 Oct, 2023 - Panelist in Bridging the Gap: Exploring the Role of AI in the Climate and Health Intersection, Johns Hopkins India Institute, Washington DC
- 6 Aug, 2023 - Panelist in American Statistical Association Graduate Student Mentoring Session at the Joint Statistical Meetings 2023, Toronto, CA
- 7 2020 - Panelist in Faculty workshop on Mentoring PhD students, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD

INVITED PRESENTATIONS

- 8 Aug 2024 (upcoming) Joint Statistical Meetings, Oregon, Portland
- 9 Nov 2023 The International Environmetrics Society Webinar Series on Interdisciplinary applications of statistics and Machine Learning for environmental applications
- 10 Nov 2023 Department of Statistics, Federal University of Minas Gerais (UFMG), Sao Paulo, Brazil (virtual)
- 11 Oct 2023 Child and Adolescent Cause of Death Estimation Meeting, World Health Organization, Geneva, Switzerland
- 12 Oct 2023 Department of Biostatistics, Yale University, New Haven, CT
- 13 Oct 2023 Department of Statistics, Cornell University, Ithaca, NY
- 14 Oct 2023 MITS Surveillance Alliance Annual Meeting, Nairobi, Kenya
- 15 Sep 2023 EnviBayes workshop, Fort Collins CO
- 16 Sep 2023 Department of Biostatistics, University of Utah, Salt Lake City, UT
- 17 Sep 2023 Department of Statistics, Brigham Young University, Provo, UT
- 18 Sep 2023 (scheduled) Department of Statistics, Virginia Tech University, Blacksburg, VA

- 19 Jul 2023 EcoSTAT Conference, Tokyo, Japan (virtual)
- 20 Jun 2023, International Chinese Statistical Association Conference, Ann Arbor, MI
- 21 Jun 2023 SIAM Conference on Optimization, Seattle, WA
- 22 Jun 2023 The Bill and Melinda Gates Foundation, Seattle, WA
- 23 May 2023 Center for Statistics and the Social Sciences, University of Washington, Seattle, WA
- 24 May 2023, Statistics, Computer Science, and Mathematics Department, The Public University of Navarre, Pamplona, Spain
- 25 May 2023, BIRS workshop on Machine Learning and Spatial Extremes, Granada, Spain
- 26 Dec 2022 International Indian Statistical Association Conference, Bengaluru, India.
- 27 Dec 2022, CMStat conference, Kings College, London, UK
- 28 Oct 2022, Data Science Conference, Texas A&M University, College Station, TX
- 29 Oct 2022, Southern Regional Council on Statistics (SRCOS) Conference, Jekyll Island, GA
- 30 Sep 2022, Department of Population, Family and Reproductive Health, Johns Hopkins University, Baltimore, MD
- 31 Aug 2022, Statistical Data Science Workshop, University of Bologna, Italy
- 32 Aug 2022 Joint Statistical Meetings, Washington DC.
- 33 Jul 2022, International Biometrics Conference (IBC2022)
- 34 May 2022 North-Eastern Statistics Symposium (NESS) (virtual)
- 35 Mar 2022 Eastern North American Region Meetings (ENAR), International Biometric Society, Houston, TX
- 36 Feb 2022, Department of Biostatistics, McGill University, Montreal, CA
- 37 Nov 2021 East Asia Chapter of the International Society for Bayesian Analysis (EAC ISBA) (virtual)
- 38 Oct 2021, Department of Statistics, Purdue University, West Lafayette, IA
- 39 Sep 2021, Department of Biostatistics, University of Minnesota, Twin Cities
- 40 Aug 2021 Joint Statistical Meetings, Seattle, WA
- 41 May 2021 Center for Disease Control (CDC) Division of Global HIV and TB's Key Population Surveillance Team (virtual)
- 42 Apr 2021 IHME-CHAMPS Convening (virtual)
- 43 Apr 2021 Department of Biostatistics, NYU School of Global Public Health, New York City, NY
- 44 Apr 2021 MCEE Cause of Death (COD) Misclassification Methods Meeting (virtual).
- 45 Mar 2021 SIAM Conference on Computational Science and Engineering (virtual)
- 46 Oct 2020, Department of Statistics, Iowa State University, Ames, IA

- 47 Oct 2020, Department of Biostatistics, Virginia Commonwealth University, Richmond, VA
- 48 Sep 2020, RTI International, Raleigh, NC
- 49 Sep 2020 MITS Surveillance Alliance Meeting (virtual)
- 50 Aug 2020 Joint Statistical Meetings (virtual)
- 51 Mar 2020 Eastern North American Region Meetings (ENAR), International Biometric Society (virtual)
- 52 Dec 2020 International Indian Statistical Association Conference, Mumbai, India.
- 53 Sept 2019, Department of Statistics, Penn State University, State College, PA
- 54 Aug 2019 International Statistical Institute World Congress, Kuala Lumpur, Malaysia.
- 55 Aug 2019 Joint Statistical Meetings, Denver, CO.
- 56 May 2019, LRI Causes and Etiologies Meeting, Baltimore, MD.
- 57 # Mar 2019 SEARCH Scientific Advisory Committee meeting, Yale University, New Haven, CT.
- 58 Mar 2019, Child Health and Mortality Prevention Surveillance (CHAMPS) program, Emory University, Atlanta, GA.
- 59 Feb 2019, Department of Biostatistics, UCLA, Los Angeles, CA
- 60 Jan 2019 Interdisciplinary Statistical Research Unit, Indian Statistical Institute, Kolkata, India.
- 61 Aug 2018 Joint Statistical Meetings, Vancouver, Canada.
- 62 Jun 2018 ISBA World Meeting, Edinburgh, UK.
- 63 Jun 2018, MITS Surveillance Alliance Inaugural Meeting, Barcelona, Spain
- 64 Mar 2018 Eastern North American Region Meetings (ENAR), International Biometric Society, Atlanta, GA.
- 65 Dec 2017 International Indian Statistical Association Conference, Hyderabad, India.
- 66 Dec 2017 10th International Conference of the ERCIM WG on Computational and Methodological Statistics, London, UK.
- 67 Nov 2017 American Public Health Association Annual Meeting, Atlanta, GA.
- 68 Oct 2017 UNAIDS Reference Group Fall Meeting 16-18 October 2017, London, UK.
- 69 Feb 2017 CDC Consultation Conference on Key Populations, CDC, Atlanta, GA.
- 70 Feb 2017 Department of Mathematics and Statistics, University of Maryland , Baltimore County, MD.
- 71 Nov 2016 President's Emergency Plan for Aids Relief, Washington DC.
- 72 Feb 2016 Department of Statistical Science, Duke University, Durham, NC.
- 73 Feb 2016 Department of Biostatistics, University of Michigan, Ann Arbor, MI.

- 74 Feb 2016 Department of Biostatistics, Johns Hopkins University, Baltimore, MD.
- 75 Feb 2016 Department of Statistics, University of California, Irvine, CA.
- 76 Feb 2016 Department of Biostatistics, University of North Carolina, Chapel Hill, NC.
- 77 Dec 2016 Platinum Jubilee International Conference on Applications of Statistics, Calcutta University, Kolkata, India.